SUPERMARKET SALES ANALYSIS

Data Analytics Project Report

Project Type: Data Analytics

Dataset: Supermarket Sales Dataset (500 Transactions)

Prepared by

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Python • Pandas • SQL • Power BI • Data Visualization

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1. Problem Statement

The aim of this project is to analyze supermarket sales data and find useful information about products, branches, categories, customers, payment methods, and customer ratings. The analysis converts raw transaction data into simple business insights that can support better sales, inventory, and customer-service decisions.

2. Dataset

The dataset contains 500 sales transactions. It includes information related to products, branches, cities, customer types, quantities, unit prices, payment methods, ratings, and sales.

Field / Area	Description
Product	Product sold by the supermarket
Branch	Branch where the transaction occurred
City	City associated with the branch
Customer Type	Member or Normal customer
Quantity	Number of units purchased
Unit Price	Price per unit
Payment Method	Method used for payment
Rating	Customer rating out of 5
Sales	Total transaction sales value

3. Objectives

- Analyze overall supermarket sales performance.
- Identify the highest-selling products and categories.
- Compare the performance of different branches.
- Analyze customer types and spending behavior.
- Identify the most popular payment method.
- Study customer ratings.
- Generate business recommendations from the results.

4. Methodology / Steps Used for Analysis

1. Collect and load the CSV dataset.

2. Check the data for missing, duplicate, or incorrect values.

3. Calculate Sales = Quantity × Unit Price.

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4. Group and summarize the data using totals, counts, and averages.

5. Create charts to compare products, branches, categories, customers, and payments.

6. Interpret the results and make business decisions.

Sales Calculation

```
Sales = Quantity × Unit Price
```


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5. Live Analysis and Key Findings

Business Question	Result
Which product generates the highest sales?	Cheese — ■27,906.30
Which branch performs best?	Branch C (Mumbai) — ■72,469.45
Which category sells the most?	Beverages — ■56,108.24
What is the most popular payment method?	UPI — 127 transactions
Do Members spend more than Normal customers?	No. Member average: ■483.14; Normal average: ■497.07
What is the average customer rating?	3.99 out of 5

5.1 Product Analysis

Cheese generated the highest sales among the products analyzed, with total sales of ■27,906.30. This indicates strong customer demand for the product and suggests that inventory planning should ensure sufficient availability.

5.2 Branch Analysis

Branch C, located in Mumbai, was the best-performing branch with sales of ■72,469.45. The supermarket can study the factors behind this performance and apply successful practices to other branches.

5.3 Category Analysis

Beverages recorded the highest category sales at ■56,108.24. Maintaining adequate stock and monitoring demand for beverage products can help avoid lost sales.

5.4 Payment Analysis

UPI was the most frequently used payment method, accounting for 127 transactions. This shows that digital payment support is important for the supermarket's customers.

5.5 Customer Analysis

Normal customers had a higher average transaction value than Members in this dataset. The average Member transaction was ■483.14, while Normal customers averaged ■497.07.

5.6 Customer Rating

The average customer rating was 3.99 out of 5. This represents a generally positive customer experience while also leaving room for improvement in service quality.

6. Business Decisions and Recommendations

- **Increase stock of high-selling products:** Keep sufficient inventory for products such as Cheese.
- **Focus on high-performing categories:** Monitor Beverages closely because they generated the highest category sales.

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- **Study Branch C:** Analyze why Branch C performs strongly and apply useful practices to other branches.
- **Support UPI payments:** Continue providing convenient digital payment options.
- **Improve customer service:** Use rating feedback to identify areas where customer satisfaction can be improved.
- **Improve membership offers:** Use customer spending patterns to design more attractive loyalty and membership benefits.

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7. Tools and Technologies

Technology	Purpose
Python	Data analysis and automation
Pandas	Data cleaning and manipulation
NumPy	Numerical operations
Matplotlib / Seaborn	Data visualization
SQL	Data querying and aggregation
Power BI	Interactive dashboard creation
Jupyter Notebook	Exploratory analysis and documentation
Git & GitHub	Version control and project sharing

8. Suggested Dashboard

- Sales Overview — total sales, transactions, quantity, and average rating.
- Branch Analysis — sales comparison by branch and city.
- Product Analysis — top products by sales and quantity.
- Category Analysis — category-wise sales comparison.
- Customer Analysis — Member vs Normal customer spending.
- Payment Analysis — transaction count by payment method.
- Rating Analysis — average customer rating and rating distribution.

9. GitHub Project Structure

```
Supermarket-Sales-Analysis/
■■■ data/
■ ■■■ supermarket_sales.csv
■■■ notebooks/
■ ■■■ supermarket_sales_analysis.ipynb
■■■ sql/
■ ■■■ sales_analysis.sql
■■■ dashboard/
■ ■■■ Supermarket_Sales_Dashboard.pbix
■■■ screenshots/
■ ■■■ sales_overview.png
■ ■■■ branch_analysis.png
■ ■■■ product_analysis.png
■ ■■■ payment_analysis.png
■■■ src/
■ ■■■ sales_analysis.py
■■■ requirements.txt
```


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```
■■■ README.md
■■■ .gitignore
```

10. Conclusion

This supermarket sales analysis demonstrates how raw transaction data can be transformed into meaningful business insights using data analytics techniques. The project identifies top-performing products, branches, categories, payment methods, customer behavior, and customer satisfaction. These insights can help a supermarket improve inventory planning, customer service, payment facilities, and overall sales performance.

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